

SPY Regime Model

HMM vs GMM • Out-of-Sample Strategy Evaluation

Autogenerated analytical report based on the supplied regime, feature, equity, fold, parameter and plot datasets.

2011–2026 Analysis window	188 Walk-forward folds	4,004 Regime observations	HMM + GMM Strategy models
-------------------------------------	----------------------------------	-------------------------------------	-------------------------------------

Executive readout

The supplied backtest shows that both regime-aware strategies underperformed passive buy-and-hold over the full evaluation window. GMM produced a higher cumulative return and higher CAGR than HMM, while HMM had the slightly higher fold hit rate. Both strategies reduced maximum drawdown relative to buy-and-hold, indicating that the regime allocation framework traded some upside participation for a different risk profile.

Important interpretation

This is a descriptive backtest report, not a recommendation or forecast. Results are based strictly on the supplied files. Transaction costs, slippage, taxes, financing effects and live execution differences are not independently added unless already embedded in the supplied data.

Generated: 20 September 2026

1. Executive Summary

207.45% HMM cumulative return	269.33% GMM cumulative return	585.75% Buy & hold cumulative	15.7 years Evaluation period
---	---	---	--

Metric	HMM	GMM	Buy & Hold
Cumulative return	207.45%	269.33%	585.75%
CAGR	7.42%	8.68%	13.05%
Sharpe ratio	0.69	0.83	0.82
Maximum drawdown	-27.17%	-20.52%	-36.98%
Final equity index	3.075	3.693	6.858

What the numbers say

HMM generated 207.45% cumulative return versus 551.01% for the matching buy-and-hold curve. GMM generated 269.33% versus 585.75%. On the supplied walk-forward folds, HMM beat buy-and-hold in 26.6% of folds, while GMM did so in 27.1% of folds.

The most notable risk characteristic is drawdown: HMM's maximum drawdown was -27.17%, GMM's was -20.52%, versus -36.98% for buy-and-hold. This indicates that the regime strategies materially changed exposure during adverse periods, although the reduction in drawdown came with lower long-run participation in the equity curve.

The fold-level results are less favorable than the aggregate equity curve alone might suggest: the mean outperformance per test fold was -0.436 percentage points for HMM and -0.370 for GMM. Both means are negative.

2. Data Coverage & Inputs

Dataset	Rows	Key fields / purpose
strategy_params.csv	5	Regime-specific position sizing, stops, targets and signal filters
SPY_features.csv	4004	5-day/21-day returns, realized volatility, SMA/EMA distances, high-volatility flag
SPY_hmm_regimes.csv	4004	Daily HMM state probabilities and mapped regimes
SPY_gmm_regimes.csv	4004	Daily GMM state probabilities and mapped regimes
SPY_hmm_folds.csv	188	188 walk-forward test folds and fold-level returns
SPY_gmm_folds.csv	188	188 walk-forward test folds and fold-level returns
SPY_hmm_equity.csv	3906	Daily HMM strategy and buy-and-hold equity curves
SPY_gmm_equity.csv	3948	Daily GMM strategy and buy-and-hold equity curves
plots.zip	3 plots	HMM dashboard, GMM dashboard and regime distribution visualization

Regime files span 2010-10-18 to 2026-09-18 (4,004 observations). The equity curves cover approximately 15.7 years of out-of-sample performance beginning 2011-01-03 and ending 2026-09-15.

The feature set is compact and interpretable. It combines short- and medium-horizon returns with realized volatility and distance from trend baselines, providing information about momentum, trend, volatility and market positioning.

3. Regime Strategy Rules

Regime	Position	Stop	Target	Signal filter	Operational interpretation
Bull	100%	5.0%	20.0%	trend_following	Full exposure, trend-following, wide stops.
Bear	30%	2.0%	8.0%	mean_reversion	Reduced exposure, tight stops, mean-reversion bias or cash.
High-Volatility	50%	3.0%	10.0%	volatility_breakout	Half exposure, volatility-breakout signals, moderate stops.
Sideways	70%	4.0%	12.0%	range_bound	Mostly invested, range-bound / oscillator-based signals.
Unknown	50%	3.0%	10.0%	neutral	Fallback: conservative half-exposure.

The parameter table defines a regime-to-exposure policy. Bull is the highest-exposure state, while Bear sharply reduces exposure. High-Volatility and Unknown use 50% exposure, and Sideways retains 70% exposure. The signal filter changes with the detected regime rather than using one universal trading rule.

A key design implication is that the strategy is not only a regime classifier. It is a portfolio-allocation layer: the same market can generate materially different strategy returns depending on the detected regime because position size, stop loss, target and signal logic change together.

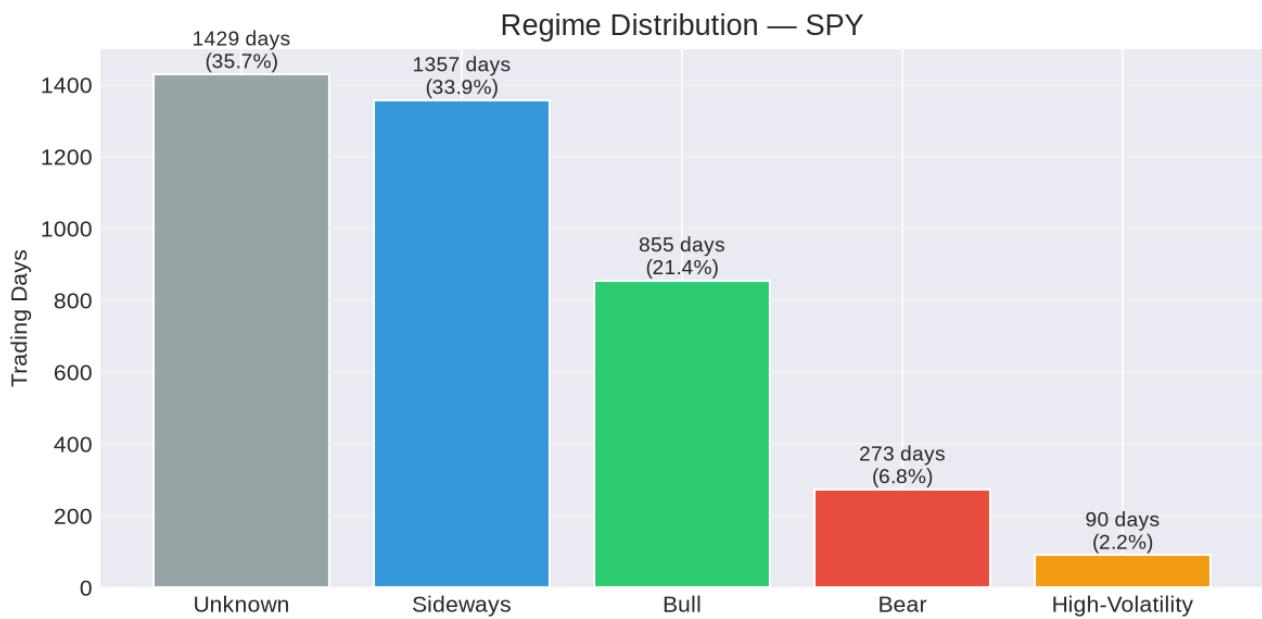
4. Feature Diagnostics

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
log_return_5d	4004	0.0027	0.0222	-0.1981	-0.0069	0.0045	0.0146	0.1601
log_return_21d	4004	0.0114	0.0429	-0.3968	-0.0072	0.0169	0.0357	0.2246
realized_vol_21d	4004	0.1447	0.0907	0.0342	0.0927	0.1226	0.1681	0.9367
sma50_dist	4004	0.0137	0.0349	-0.2740	-0.0011	0.0194	0.0361	0.1325
sma200_dist	4004	0.0554	0.0586	-0.2568	0.0304	0.0667	0.0936	0.1785
ema_ratio	4004	1.0037	0.0100	0.9152	0.9996	1.0056	1.0102	1.0306
high_vol_flag	4004	0.0907	0.2872	0.0000	0.0000	0.0000	0.0000	1.0000

Feature-level insights

- 21-day realized volatility averaged 0.145 with a maximum of 0.937, showing occasional volatility spikes.
- The high-volatility flag was active on 9.1% of observations, so the explicit high-volatility condition is relatively selective.
- The median SMA200 distance was 0.067; the median EMA ratio was 1.0056, encoding longer- and shorter-term trend position.
- The 21-day log-return median was 0.0169, with an observed range from -0.3968 to 0.2246.

5. Regime Detection & Distribution



HMM regime mix

Regime	Observations	Share
Unknown	1,709	42.68%
Sideways	1,438	35.91%
Bull	494	12.34%
Bear	274	6.84%
High-Volatility	89	2.22%

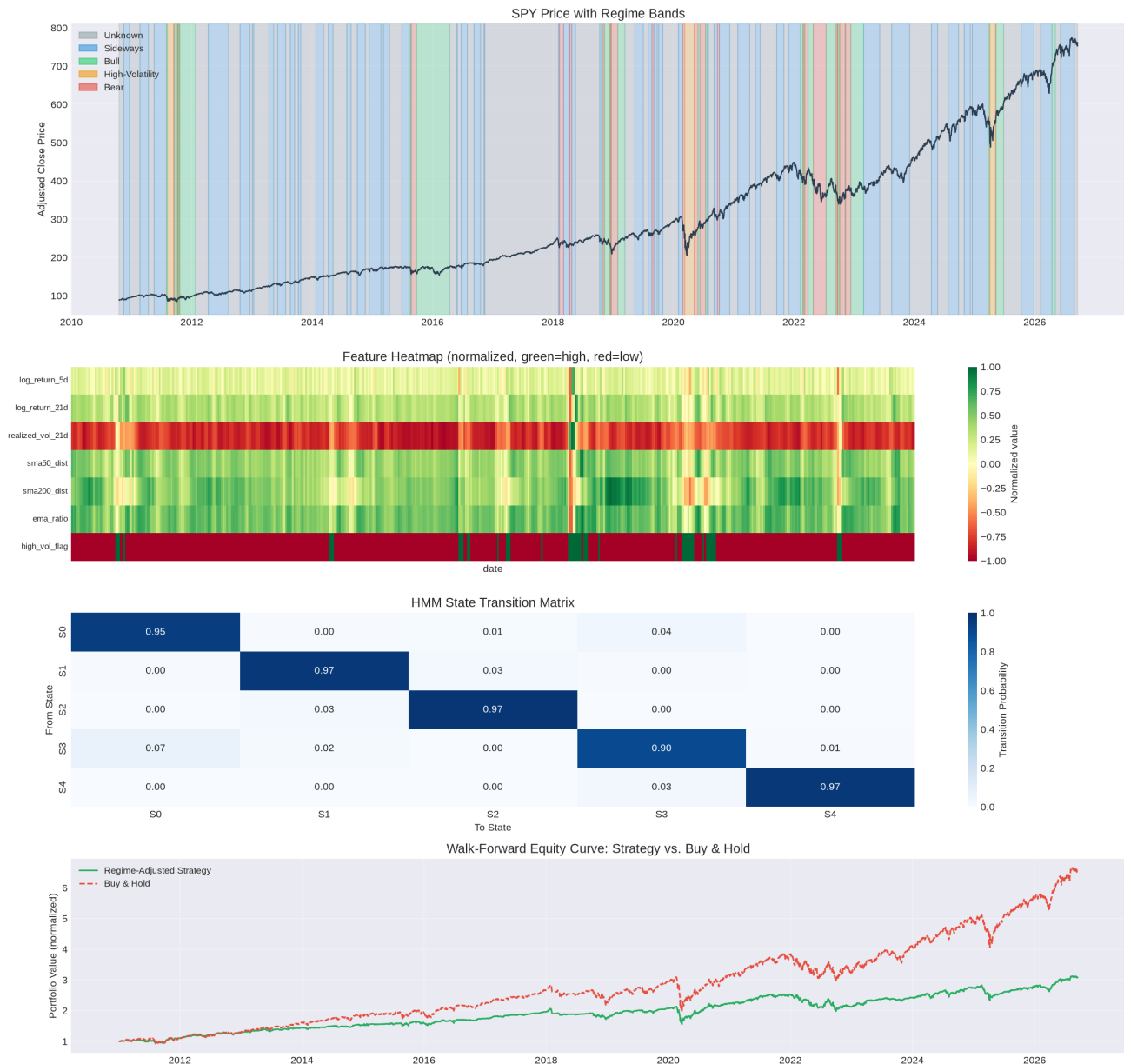
GMM regime mix

Regime	Observations	Share
Unknown	1,429	35.69%
Sideways	1,357	33.89%
Bull	855	21.35%
Bear	273	6.82%
High-Volatility	90	2.25%

The regime mapping is not symmetric between models. HMM assigns substantially more observations to Unknown, while GMM assigns more observations to Bull. Because strategy parameters differ by regime, these classification differences directly influence exposure and therefore compound returns.

6. HMM Model Analysis

Market Regime Detection — SPY [HMM]



-0.44%

Fold mean outperformance

26.6%

Positive folds

5.04%

Best fold

-6.23%

Worst fold

HMM's state probabilities are highly concentrated for several mapped regimes. In the supplied data, Bull, Sideways and High-Volatility each map strongly to distinct states, while Unknown is also concentrated. This indicates a relatively clean state-to-regime mapping in the exported probabilities.

Despite that clean classification structure, HMM's walk-forward mean outperformance was -0.436 percentage points per fold and the median was -0.268. Only 50 of 188 folds were positive relative to buy-and-hold.

7. GMM Model Analysis

Market Regime Detection — SPY [GMM]



-0.37%

Fold mean outperformance

27.1%

Positive folds

12.89%

Best fold

-8.74%

Worst fold

GMM produced a higher aggregate strategy return than HMM in the supplied equity curve. Its state mapping is also distinct: Bull is concentrated in one state, Bear in another, High-Volatility in a third, while Sideways and Unknown share probability mass across a smaller set of states.

GMM's fold-level mean outperformance was -0.370 percentage points, with a median of -0.377. It exceeded buy-and-hold in 51 of 188 folds. The model's best fold was 12.890%, but its worst was -8.739%, showing greater fold-to-fold dispersion than HMM.

8. HMM vs GMM: Direct Comparison

Measure	HMM	GMM	Difference / observation
Cumulative strategy return	207.45%	269.33%	61.88% GMM higher
CAGR	7.42%	8.68%	1.26% GMM higher
Sharpe	0.69	0.83	0.14
Max drawdown	-27.17%	-20.52%	6.66% difference
Positive folds	26.6%	27.1%	0.5 pp
Mean fold outperformance	-0.44%	-0.37%	0.07% pp
Best fold	5.04%	12.89%	GMM has larger upside outlier
Worst fold	-6.23%	-8.74%	GMM has larger downside outlier

Interpretation

Within the supplied experiment, GMM has the stronger aggregate strategy equity curve and the higher CAGR, while HMM has a marginally higher percentage of positive folds. Neither model demonstrates persistent positive average fold outperformance versus buy-and-hold. The difference between HMM and GMM should therefore be understood primarily as a difference in regime segmentation and exposure timing, not as evidence that regime detection alone creates excess return.

The risk trade-off is also visible. Both strategies reduce maximum drawdown versus buy-and-hold. GMM has the smaller maximum drawdown of the two strategies, but its fold outcomes are more dispersed: its best fold is substantially stronger and its worst fold substantially weaker than HMM's.

9. Walk-Forward Validation

The fold files contain 188 sequential train/test windows for each model. Each row compares the regime strategy's test-period return with the corresponding buy-and-hold return. This structure evaluates the strategy repeatedly on future periods rather than relying on a single aggregate in-sample result.

HMM fold distribution

Statistic	Outperformance per test fold
Mean	-0.44%
10th percentile	-2.18%
25th percentile	-1.36%
Median	-0.27%
75th percentile	0.07%
90th percentile	1.26%
Positive folds	26.6%
Standard deviation	1.50%

GMM fold distribution

Statistic	Outperformance per test fold
Mean	-0.37%

Statistic	Outperformance per test fold
10th percentile	-2.14%
25th percentile	-1.38%
Median	-0.38%
75th percentile	0.17%
90th percentile	1.63%
Positive folds	27.1%
Standard deviation	2.10%

10. Fold-Level Appendix — HMM

All 188 HMM test folds are reproduced below for auditability. Values are taken directly from the supplied CSV.

Fold	Test start	Test end	States	Strategy %	B&H %	Outperf. %	Regimes detected
0	2011-01-03	2011-02-01	5	0.873	2.904	-2.032	Bear
1	2011-02-02	2011-03-03	4	1.560	2.284	-0.723	Bear High-Volatility Bull Sideways
2	2011-03-04	2011-04-01	4	0.682	0.950	-0.268	High-Volatility Sideways
3	2011-04-04	2011-05-03	3	0.930	1.854	-0.923	High-Volatility
4	2011-05-04	2011-06-02	4	-3.740	-2.299	-1.441	High-Volatility Bull
5	2011-06-03	2011-07-01	4	2.245	3.193	-0.948	Bear Sideways
6	2011-07-05	2011-08-02	4	-4.371	-6.218	1.847	Sideways
7	2011-08-03	2011-08-31	4	-0.761	-3.131	2.370	Bear
8	2011-09-01	2011-09-30	5	-2.941	-5.957	3.016	Unknown
9	2011-10-03	2011-10-31	4	14.164	14.164	0.000	Bull Sideways
10	2011-11-01	2011-11-30	5	1.780	2.451	-0.671	Sideways
11	2011-12-01	2011-12-30	5	0.347	1.061	-0.714	Bear Unknown
12	2012-01-03	2012-02-01	5	3.898	3.898	0.000	High-Volatility Bull
13	2012-02-02	2012-03-02	5	2.436	3.490	-1.053	Sideways
14	2012-03-05	2012-04-02	4	2.918	4.177	-1.259	Bear Sideways
15	2012-04-03	2012-05-02	4	-0.523	-0.665	0.143	High-Volatility Bull Sideways
16	2012-05-03	2012-06-01	5	-4.800	-7.964	3.164	High-Volatility Sideways
17	2012-06-04	2012-07-02	5	7.117	7.117	0.000	Bull
18	2012-07-03	2012-08-01	4	0.131	0.131	0.000	Bull Sideways
19	2012-08-02	2012-08-30	3	2.818	2.818	0.000	High-Volatility Bull
20	2012-08-31	2012-10-01	5	0.844	2.806	-1.962	Bear
21	2012-10-02	2012-11-01	5	-0.567	-1.156	0.589	High-Volatility
22	2012-11-02	2012-12-03	2	-0.078	-0.078	0.000	High-Volatility Bull
23	2012-12-04	2013-01-03	4	2.734	3.903	-1.169	Bull Sideways
24	2013-01-04	2013-02-04	5	0.650	2.166	-1.516	Bear Sideways
25	2013-02-05	2013-03-06	5	2.247	2.284	-0.037	High-Volatility Bull Sideways
26	2013-03-07	2013-04-05	5	0.691	0.691	0.000	High-Volatility Bull
27	2013-04-08	2013-05-06	4	1.742	3.566	-1.824	Bear High-Volatility Sideways
28	2013-05-07	2013-06-05	4	-0.818	-0.818	0.000	Bull
29	2013-06-06	2013-07-05	4	0.011	0.708	-0.698	Bear High-Volatility Sideways
30	2013-07-08	2013-08-05	5	2.870	4.117	-1.247	Bull Sideways
31	2013-08-06	2013-09-04	5	-1.169	-2.345	1.176	High-Volatility Unknown
32	2013-09-05	2013-10-03	5	1.148	1.493	-0.345	Bear Sideways Unknown
33	2013-10-04	2013-11-01	5	2.160	4.334	-2.174	Unknown
34	2013-11-04	2013-12-03	2	0.030	1.651	-1.621	High-Volatility Bull
35	2013-12-04	2014-01-03	5	0.693	2.311	-1.618	Bear Unknown
36	2014-01-06	2014-02-04	5	-2.673	-3.822	1.149	Sideways Unknown
37	2014-02-05	2014-03-06	4	3.659	7.427	-3.768	High-Volatility
38	2014-03-07	2014-04-04	4	-0.378	-0.551	0.173	Sideways
39	2014-04-07	2014-05-06	5	0.674	1.324	-0.650	High-Volatility Sideways Unknown
40	2014-05-07	2014-06-05	3	2.440	3.497	-1.057	Sideways
41	2014-06-06	2014-07-07	4	0.788	1.575	-0.787	High-Volatility

Fold	Test start	Test end	States	Strategy %	B&H %	Outperf. %	Regimes detected
42	2014-07-08	2014-08-05	2	-2.518	-2.156	-0.362	High-Volatility Bull
43	2014-08-06	2014-09-04	4	1.991	4.238	-2.247	Bear High-Volatility Sideways
44	2014-09-05	2014-10-03	2	-0.758	-1.826	1.067	Bear Bull
45	2014-10-06	2014-11-03	2	2.156	2.792	-0.636	Bear Bull
46	2014-11-04	2014-12-03	5	2.365	3.392	-1.027	Sideways
47	2014-12-04	2015-01-05	3	-1.730	-2.324	0.594	Bear High-Volatility Bull
48	2015-01-06	2015-02-04	5	2.122	2.122	0.000	Bull
49	2015-02-05	2015-03-06	4	0.207	0.669	-0.463	Bear High-Volatility
50	2015-03-09	2015-04-07	3	-0.007	-0.074	0.067	Bear
51	2015-04-08	2015-05-06	3	0.017	0.029	-0.012	Bear
52	2015-05-07	2015-06-05	4	0.224	0.431	-0.207	High-Volatility Bull
53	2015-06-08	2015-07-07	2	0.265	0.265	0.000	Bull
54	2015-07-08	2015-08-05	5	1.355	2.709	-1.353	High-Volatility
55	2015-08-06	2015-09-03	5	-3.072	-6.144	3.072	Sideways Unknown
56	2015-09-04	2015-10-05	5	3.589	3.589	0.000	Bull Unknown
57	2015-10-06	2015-11-03	5	3.301	6.679	-3.378	Unknown
58	2015-11-04	2015-12-03	4	-1.572	-2.258	0.686	Bear Sideways
59	2015-12-04	2016-01-05	4	-2.348	-3.369	1.021	High-Volatility Sideways
60	2016-01-06	2016-02-04	5	-0.882	-3.631	2.749	Bear Bull Unknown
61	2016-02-05	2016-03-07	4	3.621	6.725	-3.105	Bear High-Volatility Sideways
62	2016-03-08	2016-04-06	3	1.461	4.579	-3.118	Bear High-Volatility Bull
63	2016-04-07	2016-05-05	4	0.580	0.500	0.080	Bull Sideways
64	2016-05-06	2016-06-06	2	1.370	2.737	-1.367	High-Volatility Bull
65	2016-06-07	2016-07-06	2	-0.439	-0.439	0.000	Bull
66	2016-07-07	2016-08-04	5	2.006	3.284	-1.277	Sideways Unknown
67	2016-08-05	2016-09-02	5	-0.037	0.087	-0.125	Bull Sideways
68	2016-09-06	2016-10-04	5	-0.728	-1.491	0.763	Bull Unknown
69	2016-10-05	2016-11-02	5	-1.370	-2.732	1.361	Bull Unknown
70	2016-11-03	2016-12-02	4	3.636	5.221	-1.585	Sideways
71	2016-12-05	2017-01-04	4	0.935	3.129	-2.194	Bear
72	2017-01-05	2017-02-03	5	1.299	1.299	0.000	Bull
73	2017-02-06	2017-03-07	4	2.459	3.525	-1.066	Sideways
74	2017-03-08	2017-04-05	3	-0.156	-0.321	0.165	High-Volatility Bull
75	2017-04-06	2017-05-05	4	1.841	1.809	0.031	High-Volatility Bull Sideways
76	2017-05-08	2017-06-06	5	0.745	1.481	-0.736	High-Volatility Unknown
77	2017-06-07	2017-07-06	5	-0.289	-0.795	0.506	Bear High-Volatility Sideways
78	2017-07-07	2017-08-04	4	1.090	2.189	-1.099	High-Volatility
79	2017-08-07	2017-09-05	2	0.032	-0.730	0.762	High-Volatility Bull
80	2017-09-06	2017-10-04	5	1.842	3.044	-1.203	High-Volatility Bull
81	2017-10-05	2017-11-02	5	0.599	1.151	-0.552	Bear Sideways Unknown
82	2017-11-03	2017-12-04	5	1.099	2.202	-1.102	High-Volatility Unknown
83	2017-12-05	2018-01-04	5	1.850	3.727	-1.876	Unknown
84	2018-01-05	2018-02-05	5	-3.524	-3.471	-0.053	Bull Sideways Unknown
85	2018-02-06	2018-03-07	0	0.000	0.000	0.000	nan
86	2018-03-08	2018-04-06	5	-2.413	-4.867	2.454	Unknown

Fold	Test start	Test end	States	Strategy %	B&H %	Outperf. %	Regimes detected
87	2018-04-09	2018-05-07	4	1.143	2.268	-1.125	High-Volatility Sideways
88	2018-05-08	2018-06-06	0	0.000	0.000	0.000	nan
89	2018-06-07	2018-07-06	5	-0.071	-0.257	0.186	Bear
90	2018-07-09	2018-08-06	3	0.799	2.425	-1.627	Bear High-Volatility Bull
91	2018-08-07	2018-09-05	5	0.365	1.208	-0.843	Bear
92	2018-09-06	2018-10-04	4	0.452	0.899	-0.447	High-Volatility Sideways
93	2018-10-05	2018-11-02	4	-3.861	-5.535	1.674	Sideways
94	2018-11-05	2018-12-04	4	-0.765	-1.149	0.384	Sideways
95	2018-12-06	2019-01-07	4	-1.665	-5.179	3.514	Bear High-Volatility Sideways
96	2019-01-08	2019-02-06	4	4.327	6.220	-1.893	Sideways
97	2019-02-07	2019-03-08	2	1.599	1.599	0.000	High-Volatility Bull
98	2019-03-11	2019-04-08	4	4.174	4.174	0.000	Bull
99	2019-04-09	2019-05-08	3	0.268	0.077	0.192	Bear High-Volatility Bull
100	2019-05-09	2019-06-07	4	0.200	0.345	-0.145	Bear High-Volatility
101	2019-06-10	2019-07-09	3	0.993	3.345	-2.352	Bear High-Volatility Bull
102	2019-07-10	2019-08-07	5	-1.777	-3.563	1.786	High-Volatility Unknown
103	2019-08-08	2019-09-06	5	0.792	1.509	-0.717	High-Volatility
104	2019-09-09	2019-10-07	3	-1.972	-1.263	-0.709	Bear Bull
105	2019-10-08	2019-11-05	4	4.453	6.412	-1.959	Sideways
106	2019-11-06	2019-12-05	3	0.482	1.602	-1.120	Bear Bull
107	2019-12-06	2020-01-07	4	1.494	3.000	-1.506	High-Volatility
108	2020-01-08	2020-02-06	5	1.476	2.937	-1.461	High-Volatility Bull
109	2020-02-07	2020-03-09	4	-12.406	-17.450	5.044	Sideways
110	2020-03-10	2020-04-07	4	-4.725	-7.534	2.809	Sideways
111	2020-04-08	2020-05-07	4	3.315	4.981	-1.666	High-Volatility Bull Sideways
112	2020-05-08	2020-06-08	4	10.503	10.518	-0.016	Bear Bull Sideways
113	2020-06-09	2020-07-08	5	-2.153	-1.003	-1.149	Bear High-Volatility Bull Sideways
114	2020-07-09	2020-08-06	4	6.346	6.346	0.000	High-Volatility Bull
115	2020-08-07	2020-09-04	4	2.391	2.391	0.000	Bull
116	2020-09-08	2020-10-06	4	0.919	0.919	0.000	Bull
117	2020-10-07	2020-11-04	4	0.281	0.816	-0.535	Bear Sideways
118	2020-11-05	2020-12-04	2	2.779	5.599	-2.820	High-Volatility
119	2020-12-07	2021-01-06	4	0.826	1.640	-0.814	Bear High-Volatility
120	2021-01-07	2021-02-05	5	1.156	2.271	-1.115	Bear Unknown
121	2021-02-08	2021-03-09	5	-0.399	-0.855	0.456	High-Volatility
122	2021-03-10	2021-04-08	5	2.585	5.205	-2.620	Unknown
123	2021-04-09	2021-05-07	3	2.583	2.583	0.000	Bull
124	2021-05-10	2021-06-08	2	1.038	1.038	0.000	Bull
125	2021-06-09	2021-07-08	2	2.533	2.533	0.000	Bull
126	2021-07-09	2021-08-06	3	0.808	1.600	-0.792	High-Volatility
127	2021-08-09	2021-09-07	5	1.478	2.110	-0.632	Bull Sideways
128	2021-09-08	2021-10-06	4	-1.874	-3.242	1.367	Bull Sideways
129	2021-10-07	2021-11-04	3	1.898	6.440	-4.542	Bear Bull
130	2021-11-05	2021-12-06	4	-0.857	-2.079	1.222	Bear Bull Sideways
131	2021-12-07	2022-01-05	3	0.105	0.373	-0.268	Bear High-Volatility

Fold	Test start	Test end	States	Strategy %	B&H %	Outperf. %	Regimes detected
132	2022-01-06	2022-02-04	5	-1.235	-4.112	2.877	Bull Unknown
133	2022-02-07	2022-03-08	5	-6.933	-6.933	0.000	Bull
134	2022-03-09	2022-04-06	5	2.279	4.796	-2.517	Bear High-Volatility
135	2022-04-07	2022-05-06	5	-7.747	-8.341	0.593	Bear Bull
136	2022-05-09	2022-06-07	3	2.482	4.413	-1.930	Bear High-Volatility
137	2022-06-08	2022-07-08	4	-4.306	-5.075	0.769	Bear High-Volatility Sideways
138	2022-07-11	2022-08-08	4	7.485	7.485	0.000	Bull Sideways
139	2022-08-09	2022-09-07	4	-3.299	-3.299	0.000	High-Volatility Bull
140	2022-09-08	2022-10-06	2	-6.406	-6.406	0.000	High-Volatility Bull
141	2022-10-07	2022-11-04	4	1.914	3.738	-1.824	High-Volatility
142	2022-11-07	2022-12-06	5	1.882	3.653	-1.771	High-Volatility Bull
143	2022-12-07	2023-01-06	2	-0.215	-0.839	0.623	Bear High-Volatility
144	2023-01-09	2023-02-07	5	4.497	7.046	-2.550	Sideways Unknown
145	2023-02-08	2023-03-09	4	-2.329	-4.649	2.320	High-Volatility
146	2023-03-10	2023-04-10	5	6.546	6.546	0.000	Bull
147	2023-04-11	2023-05-09	5	0.167	0.295	-0.128	Sideways Unknown
148	2023-05-10	2023-06-08	2	1.179	3.943	-2.765	Bear
149	2023-06-09	2023-07-11	3	0.989	3.304	-2.315	Bear
150	2023-07-12	2023-08-09	5	-0.021	-0.061	0.039	Sideways Unknown
151	2023-08-10	2023-09-08	4	-0.015	-0.087	0.072	Bear
152	2023-09-11	2023-10-09	3	-0.802	-3.263	2.461	Bear High-Volatility
153	2023-10-10	2023-11-07	3	0.159	0.550	-0.391	Bear High-Volatility
154	2023-11-08	2023-12-07	2	1.426	4.798	-3.373	Bear
155	2023-12-08	2024-01-09	4	1.015	3.390	-2.375	Bear
156	2024-01-10	2024-02-08	5	2.271	4.566	-2.295	High-Volatility
157	2024-02-09	2024-03-11	4	1.416	2.011	-0.595	Bull Sideways
158	2024-03-12	2024-04-10	4	-0.096	-0.206	0.110	Bear Bull Sideways
159	2024-04-11	2024-05-09	5	0.419	0.419	0.000	Bull
160	2024-05-10	2024-06-10	3	2.845	2.845	0.000	Bull
161	2024-06-11	2024-07-11	5	1.972	3.972	-1.999	High-Volatility Unknown
162	2024-07-12	2024-08-09	4	-4.821	-4.821	0.000	High-Volatility Bull
163	2024-08-12	2024-09-10	5	1.468	2.910	-1.442	Unknown
164	2024-09-11	2024-10-09	4	3.807	4.417	-0.611	Bear Bull Sideways
165	2024-10-10	2024-11-07	5	2.371	3.381	-1.010	Sideways
166	2024-11-08	2024-12-09	5	1.376	1.085	0.291	Bear Sideways Unknown
167	2024-12-10	2025-01-10	3	-1.003	-3.377	2.374	Bear
168	2025-01-13	2025-02-11	5	2.060	4.114	-2.054	High-Volatility Sideways
169	2025-02-12	2025-03-13	3	-8.608	-8.608	0.000	Bull
170	2025-03-14	2025-04-11	3	-3.214	-4.844	1.631	Sideways
171	2025-04-14	2025-05-13	3	2.621	8.851	-6.231	Bear
172	2025-05-14	2025-06-12	5	1.380	2.750	-1.370	Unknown
173	2025-06-13	2025-07-15	5	2.046	4.519	-2.473	Unknown Bull High-Volatility Sideways Bear
174	2025-07-16	2025-08-13	5	2.260	3.311	-1.052	Bear High-Volatility Sideways Unknown
175	2025-08-14	2025-09-12	4	0.488	1.932	-1.444	Bear Bull Sideways
176	2025-09-15	2025-10-13	2	0.600	0.600	0.000	High-Volatility Bull

Fold	Test start	Test end	States	Strategy %	B&H %	Outperf. %	Regimes detected
177	2025-10-14	2025-11-11	4	1.680	3.136	-1.457	Bear High-Volatility
178	2025-11-12	2025-12-11	5	0.439	0.847	-0.409	High-Volatility Unknown
179	2025-12-12	2026-01-13	5	2.062	2.062	0.000	Bull
180	2026-01-14	2026-02-12	5	-2.029	-1.317	-0.712	Bull Unknown
181	2026-02-13	2026-03-16	5	-0.923	-1.866	0.943	Bear High-Volatility Unknown
182	2026-03-17	2026-04-15	3	3.253	4.631	-1.378	Sideways
183	2026-04-16	2026-05-14	5	4.606	6.629	-2.023	Sideways
184	2026-05-15	2026-06-15	5	2.286	2.119	0.168	Bull Sideways
185	2026-06-16	2026-07-16	5	0.170	0.310	-0.140	High-Volatility Sideways Unknown
186	2026-07-17	2026-08-14	4	3.174	4.446	-1.272	Bear Bull Sideways
187	2026-08-17	2026-09-15	5	-1.382	-1.978	0.595	Sideways Unknown

11. Fold-Level Appendix — GMM

All 188 GMM test folds are reproduced below for auditability. Values are taken directly from the supplied CSV.

Fold	Test start	Test end	States	Strategy %	B&H %	Outperf. %	Regimes detected
0	2011-01-03	2011-02-01	5	1.443	2.904	-1.462	Sideways Unknown
1	2011-02-02	2011-03-03	5	1.153	2.284	-1.131	Unknown
2	2011-03-04	2011-04-01	5	0.495	0.950	-0.455	High-Volatility Unknown
3	2011-04-04	2011-05-03	2	2.774	1.854	0.921	Bear Bull
4	2011-05-04	2011-06-02	5	-2.299	-2.299	0.000	Bull Unknown
5	2011-06-03	2011-07-01	2	3.193	3.193	0.000	Bull
6	2011-07-05	2011-08-02	2	-3.652	-6.218	2.566	Bear Bull
7	2011-08-03	2011-08-31	4	-11.829	-3.131	-8.698	High-Volatility Bull
8	2011-09-01	2011-09-30	3	-2.918	-5.957	3.039	Bear High-Volatility
9	2011-10-03	2011-10-31	4	5.425	14.164	-8.739	Bear Sideways
10	2011-11-01	2011-11-30	4	-1.672	2.451	-4.123	Bear Bull
11	2011-12-01	2011-12-30	2	0.347	1.061	-0.714	Bear
12	2012-01-03	2012-02-01	2	1.159	3.898	-2.739	Bear
13	2012-02-02	2012-03-02	2	2.376	3.490	-1.114	High-Volatility Bull
14	2012-03-05	2012-04-02	2	4.177	4.177	0.000	Bull
15	2012-04-03	2012-05-02	5	-0.314	-0.665	0.351	High-Volatility Unknown
16	2012-05-03	2012-06-01	5	-2.437	-7.964	5.527	Bear
17	2012-06-04	2012-07-02	4	4.038	7.117	-3.080	Bear High-Volatility
18	2012-07-03	2012-08-01	5	0.175	0.131	0.044	Bear Unknown
19	2012-08-02	2012-08-30	5	1.174	2.818	-1.644	High-Volatility Bull
20	2012-08-31	2012-10-01	5	1.406	2.806	-1.400	Unknown
21	2012-10-02	2012-11-01	5	-0.567	-1.156	0.589	Unknown
22	2012-11-02	2012-12-03	4	1.621	-0.078	1.699	Bear High-Volatility Sideways
23	2012-12-04	2013-01-03	5	1.918	3.903	-1.986	Bear Unknown
24	2013-01-04	2013-02-04	3	2.166	2.166	0.000	Bull
25	2013-02-05	2013-03-06	5	1.605	2.284	-0.679	Sideways
26	2013-03-07	2013-04-05	5	0.352	0.691	-0.339	High-Volatility Unknown
27	2013-04-08	2013-05-06	5	2.510	3.566	-1.056	Bear Sideways Unknown
28	2013-05-07	2013-06-05	2	-0.818	-0.818	0.000	Bull
29	2013-06-06	2013-07-05	5	0.518	0.708	-0.191	Sideways
30	2013-07-08	2013-08-05	2	2.596	4.117	-1.521	Bear Bull
31	2013-08-06	2013-09-04	3	-1.169	-2.345	1.176	High-Volatility
32	2013-09-05	2013-10-03	4	0.644	1.493	-0.848	High-Volatility Sideways
33	2013-10-04	2013-11-01	3	4.334	4.334	0.000	Bull
34	2013-11-04	2013-12-03	2	1.651	1.651	0.000	Bull
35	2013-12-04	2014-01-03	5	0.608	2.311	-1.703	Bear Unknown
36	2014-01-06	2014-02-04	5	-2.673	-3.822	1.149	Sideways
37	2014-02-05	2014-03-06	4	5.153	7.427	-2.274	Sideways
38	2014-03-07	2014-04-04	5	-0.406	-0.551	0.145	Bull Unknown
39	2014-04-07	2014-05-06	5	0.674	1.324	-0.650	Unknown
40	2014-05-07	2014-06-05	5	1.748	3.497	-1.749	High-Volatility Sideways
41	2014-06-06	2014-07-07	5	0.788	1.575	-0.787	High-Volatility Unknown

Fold	Test start	Test end	States	Strategy %	B&H %	Outperf. %	Regimes detected
42	2014-07-08	2014-08-05	3	-0.835	-2.156	1.320	Bear High-Volatility
43	2014-08-06	2014-09-04	3	3.466	4.238	-0.772	High-Volatility Bull
44	2014-09-05	2014-10-03	2	-0.758	-1.826	1.067	Bear Bull
45	2014-10-06	2014-11-03	3	1.700	2.792	-1.092	Bear Bull
46	2014-11-04	2014-12-03	5	2.017	3.392	-1.375	Sideways Unknown
47	2014-12-04	2015-01-05	4	-0.058	-2.324	2.266	Bear Bull High-Volatility Sideways
48	2015-01-06	2015-02-04	5	1.733	2.122	-0.389	High-Volatility Sideways Unknown
49	2015-02-05	2015-03-06	4	-0.084	0.669	-0.753	Bear Sideways
50	2015-03-09	2015-04-07	4	-0.659	-0.074	-0.585	Bear Sideways
51	2015-04-08	2015-05-06	2	-0.164	0.029	-0.193	Bear Bull
52	2015-05-07	2015-06-05	2	0.308	0.431	-0.122	Sideways
53	2015-06-08	2015-07-07	5	0.196	0.265	-0.068	Sideways
54	2015-07-08	2015-08-05	3	2.709	2.709	0.000	Bull
55	2015-08-06	2015-09-03	3	-3.430	-6.144	2.713	Bear Bull
56	2015-09-04	2015-10-05	5	2.421	3.589	-1.168	Bear High-Volatility Sideways
57	2015-10-06	2015-11-03	5	3.301	6.679	-3.378	Unknown
58	2015-11-04	2015-12-03	5	-1.542	-2.258	0.716	Sideways Unknown
59	2015-12-04	2016-01-05	4	-2.348	-3.369	1.021	Sideways
60	2016-01-06	2016-02-04	4	-0.695	-3.631	2.936	Bear High-Volatility
61	2016-02-05	2016-03-07	2	2.472	6.725	-4.253	Bear High-Volatility
62	2016-03-08	2016-04-06	3	1.689	4.579	-2.890	Bear High-Volatility
63	2016-04-07	2016-05-05	3	0.500	0.500	0.000	Bull
64	2016-05-06	2016-06-06	2	1.370	2.737	-1.367	High-Volatility
65	2016-06-07	2016-07-06	4	-0.722	-0.439	-0.283	Bear High-Volatility Sideways
66	2016-07-07	2016-08-04	5	1.634	3.284	-1.649	High-Volatility Unknown
67	2016-08-05	2016-09-02	4	0.046	0.087	-0.041	High-Volatility
68	2016-09-06	2016-10-04	5	-0.728	-1.491	0.763	Unknown
69	2016-10-05	2016-11-02	3	-1.168	-2.732	1.564	Bear High-Volatility
70	2016-11-03	2016-12-02	5	5.565	5.221	0.344	Bear Bull Unknown
71	2016-12-05	2017-01-04	3	1.720	3.129	-1.409	High-Volatility Bull
72	2017-01-05	2017-02-03	5	0.618	1.299	-0.680	Bear High-Volatility
73	2017-02-06	2017-03-07	4	2.459	3.525	-1.066	Sideways
74	2017-03-08	2017-04-05	5	-0.326	-0.321	-0.005	Sideways Unknown
75	2017-04-06	2017-05-05	5	1.654	1.809	-0.156	High-Volatility Bull Sideways
76	2017-05-08	2017-06-06	4	0.954	1.481	-0.527	Bull Sideways
77	2017-06-07	2017-07-06	5	-0.552	-0.795	0.243	Sideways
78	2017-07-07	2017-08-04	3	1.092	2.189	-1.097	Bear High-Volatility
79	2017-08-07	2017-09-05	2	-0.481	-0.730	0.250	High-Volatility Bull
80	2017-09-06	2017-10-04	5	1.994	3.044	-1.050	Bear High-Volatility Sideways Unknown
81	2017-10-05	2017-11-02	4	1.151	1.151	0.000	Bull
82	2017-11-03	2017-12-04	3	1.099	2.202	-1.102	High-Volatility
83	2017-12-05	2018-01-04	4	2.598	3.727	-1.129	Sideways
84	2018-01-05	2018-02-05	2	-1.364	-3.471	2.107	High-Volatility Bull
85	2018-02-06	2018-03-07	2	0.125	1.356	-1.231	High-Volatility Bull
86	2018-03-08	2018-04-06	3	-2.413	-4.867	2.454	High-Volatility

Fold	Test start	Test end	States	Strategy %	B&H %	Outperf. %	Regimes detected
87	2018-04-09	2018-05-07	5	1.555	2.268	-0.713	Bull Unknown
88	2018-05-08	2018-06-06	5	1.955	3.926	-1.971	Unknown
89	2018-06-07	2018-07-06	5	-0.122	-0.257	0.135	Unknown
90	2018-07-09	2018-08-06	3	1.287	2.425	-1.138	Bear High-Volatility
91	2018-08-07	2018-09-05	2	1.208	1.208	0.000	Bull
92	2018-09-06	2018-10-04	2	0.452	0.899	-0.447	High-Volatility
93	2018-10-05	2018-11-02	2	-3.920	-5.535	1.615	High-Volatility Bull
94	2018-11-05	2018-12-04	5	0.417	-1.149	1.565	Bear High-Volatility Unknown
95	2018-12-06	2019-01-07	5	-3.510	-5.179	1.668	Bear High-Volatility Sideways
96	2019-01-08	2019-02-06	3	6.220	6.220	0.000	Bear Bull
97	2019-02-07	2019-03-08	4	1.122	1.599	-0.477	Sideways
98	2019-03-11	2019-04-08	5	2.077	4.174	-2.097	Unknown
99	2019-04-09	2019-05-08	4	0.060	0.077	-0.016	Sideways
100	2019-05-09	2019-06-07	2	0.345	0.345	0.000	Bull
101	2019-06-10	2019-07-09	5	1.718	3.345	-1.627	High-Volatility Sideways
102	2019-07-10	2019-08-07	4	-1.586	-3.563	1.978	High-Volatility Bull
103	2019-08-08	2019-09-06	4	-0.222	1.509	-1.731	Bear High-Volatility
104	2019-09-09	2019-10-07	2	-1.598	-1.263	-0.335	Bear Bull
105	2019-10-08	2019-11-05	3	3.164	6.412	-3.248	High-Volatility
106	2019-11-06	2019-12-05	4	1.602	1.602	0.000	Bull
107	2019-12-06	2020-01-07	4	1.525	3.000	-1.476	High-Volatility Bull
108	2020-01-08	2020-02-06	4	1.352	2.937	-1.586	Bear Sideways
109	2020-02-07	2020-03-09	5	-10.195	-17.450	7.255	High-Volatility Sideways Unknown
110	2020-03-10	2020-04-07	5	5.356	-7.534	12.890	Bear High-Volatility Sideways
111	2020-04-08	2020-05-07	3	2.541	4.981	-2.440	High-Volatility
112	2020-05-08	2020-06-08	4	7.362	10.518	-3.156	Bear Sideways
113	2020-06-09	2020-07-08	5	-0.636	-1.003	0.367	Sideways Unknown
114	2020-07-09	2020-08-06	5	4.414	6.346	-1.932	Sideways
115	2020-08-07	2020-09-04	5	0.801	2.391	-1.590	Bull Unknown
116	2020-09-08	2020-10-06	5	0.676	0.919	-0.243	Sideways
117	2020-10-07	2020-11-04	3	0.451	0.816	-0.365	High-Volatility
118	2020-11-05	2020-12-04	3	5.160	5.599	-0.439	High-Volatility Bull
119	2020-12-07	2021-01-06	3	0.973	1.640	-0.667	High-Volatility Bull
120	2021-01-07	2021-02-05	5	1.156	2.271	-1.115	Unknown
121	2021-02-08	2021-03-09	5	-0.399	-0.855	0.456	Unknown
122	2021-03-10	2021-04-08	5	3.212	5.205	-1.993	High-Volatility Sideways Unknown
123	2021-04-09	2021-05-07	2	2.537	2.583	-0.047	High-Volatility Bull
124	2021-05-10	2021-06-08	3	1.038	1.038	0.000	Bear Bull
125	2021-06-09	2021-07-08	4	1.472	2.533	-1.060	High-Volatility Sideways
126	2021-07-09	2021-08-06	5	1.127	1.600	-0.473	Sideways
127	2021-08-09	2021-09-07	3	1.056	2.110	-1.054	High-Volatility
128	2021-09-08	2021-10-06	2	-1.612	-3.242	1.630	High-Volatility Bull
129	2021-10-07	2021-11-04	5	4.640	6.440	-1.800	Sideways Unknown
130	2021-11-05	2021-12-06	3	-0.476	-2.079	1.603	Bear Bull
131	2021-12-07	2022-01-05	2	-1.875	0.373	-2.248	Bear Bull

Fold	Test start	Test end	States	Strategy %	B&H %	Outperf. %	Regimes detected
132	2022-01-06	2022-02-04	2	-4.112	-4.112	0.000	Bull
133	2022-02-07	2022-03-08	3	-2.086	-6.933	4.847	Bear
134	2022-03-09	2022-04-06	5	2.969	4.796	-1.827	Bear High-Volatility Bull
135	2022-04-07	2022-05-06	4	-3.701	-8.341	4.639	Bear High-Volatility
136	2022-05-09	2022-06-07	5	4.020	4.413	-0.392	Bull Sideways
137	2022-06-08	2022-07-08	3	-2.237	-5.075	2.839	Bear High-Volatility
138	2022-07-11	2022-08-08	3	2.391	7.485	-5.094	Bear Bull
139	2022-08-09	2022-09-07	4	-4.376	-3.299	-1.077	Bull Sideways
140	2022-09-08	2022-10-06	5	-2.909	-6.406	3.497	Bear Unknown
141	2022-10-07	2022-11-04	2	4.469	3.738	0.731	High-Volatility Bull
142	2022-11-07	2022-12-06	2	1.759	3.653	-1.894	Bear High-Volatility
143	2022-12-07	2023-01-06	2	-0.384	-0.839	0.455	High-Volatility
144	2023-01-09	2023-02-07	4	4.695	7.046	-2.351	High-Volatility Bull
145	2023-02-08	2023-03-09	2	-4.649	-4.649	0.000	Bull
146	2023-03-10	2023-04-10	2	6.546	6.546	0.000	Bull
147	2023-04-11	2023-05-09	3	0.642	0.295	0.347	High-Volatility Bull
148	2023-05-10	2023-06-08	2	0.732	3.943	-3.212	Bear Bull
149	2023-06-09	2023-07-11	2	1.649	3.304	-1.655	High-Volatility
150	2023-07-12	2023-08-09	3	-0.479	-0.061	-0.419	Bear Bull
151	2023-08-10	2023-09-08	3	-0.031	-0.087	0.057	High-Volatility
152	2023-09-11	2023-10-09	3	-1.203	-3.263	2.061	High-Volatility Bull
153	2023-10-10	2023-11-07	5	1.287	0.550	0.737	High-Volatility Bull
154	2023-11-08	2023-12-07	2	1.426	4.798	-3.373	Bear
155	2023-12-08	2024-01-09	3	1.186	3.390	-2.204	Bear High-Volatility
156	2024-01-10	2024-02-08	4	3.186	4.566	-1.380	Sideways
157	2024-02-09	2024-03-11	2	1.015	2.011	-0.996	High-Volatility
158	2024-03-12	2024-04-10	3	-0.206	-0.206	0.000	Bull
159	2024-04-11	2024-05-09	5	0.759	0.419	0.341	Bear High-Volatility
160	2024-05-10	2024-06-10	2	2.845	2.845	0.000	Bear Bull
161	2024-06-11	2024-07-11	5	3.513	3.972	-0.458	Bull Sideways
162	2024-07-12	2024-08-09	4	-3.194	-4.821	1.628	Bear Bull Sideways
163	2024-08-12	2024-09-10	4	2.546	2.910	-0.364	Bull Sideways
164	2024-09-11	2024-10-09	5	4.417	4.417	0.000	Bull
165	2024-10-10	2024-11-07	4	2.462	3.381	-0.919	Bull Sideways
166	2024-11-08	2024-12-09	5	0.527	1.085	-0.557	Bear Unknown
167	2024-12-10	2025-01-10	3	-2.579	-3.377	0.798	High-Volatility Bull
168	2025-01-13	2025-02-11	4	2.286	4.114	-1.828	Bear Bull
169	2025-02-12	2025-03-13	4	-8.608	-8.608	0.000	Bull
170	2025-03-14	2025-04-11	4	-6.918	-4.844	-2.073	Bull Sideways
171	2025-04-14	2025-05-13	5	6.732	8.851	-2.119	High-Volatility Bull
172	2025-05-14	2025-06-12	5	1.380	2.750	-1.370	Unknown
173	2025-06-13	2025-07-15	5	2.244	4.519	-2.275	Unknown
174	2025-07-16	2025-08-13	5	1.654	3.311	-1.657	Unknown
175	2025-08-14	2025-09-12	5	0.582	1.932	-1.350	Bear
176	2025-09-15	2025-10-13	3	-0.528	0.600	-1.129	Bear High-Volatility Bull

Fold	Test start	Test end	States	Strategy %	B&H %	Outperf. %	Regimes detected
177	2025-10-14	2025-11-11	4	3.136	3.136	0.000	Bull
178	2025-11-12	2025-12-11	4	0.030	0.847	-0.817	Bear High-Volatility
179	2025-12-12	2026-01-13	2	2.062	2.062	0.000	Bull
180	2026-01-14	2026-02-12	2	-1.317	-1.317	0.000	Bull
181	2026-02-13	2026-03-16	3	-0.923	-1.866	0.943	High-Volatility
182	2026-03-17	2026-04-15	2	5.155	4.631	0.524	Bear Bull
183	2026-04-16	2026-05-14	4	4.982	6.629	-1.647	Bull Sideways
184	2026-05-15	2026-06-15	5	1.030	2.119	-1.088	Sideways Unknown
185	2026-06-16	2026-07-16	5	0.170	0.310	-0.140	Unknown
186	2026-07-17	2026-08-14	5	4.446	4.446	0.000	Bull
187	2026-08-17	2026-09-15	3	-1.978	-1.978	0.000	Bull

12. Key Insights & Analytical Caveats

Key insights

- The regime framework changes exposure meaningfully across market states, rather than merely labeling the market.
- Both models underperformed buy-and-hold on cumulative return: HMM by 378.30 percentage points and GMM by 316.42 percentage points.
- GMM's supplied strategy equity curve finished above HMM's (3.693 vs 3.075), but both remained below their corresponding buy-and-hold curves.
- The strongest evidence of risk-management value is the lower maximum drawdown of the regime strategies compared with buy-and-hold.
- Fold-level consistency remains a limitation: only 26.6% of HMM folds and 27.1% of GMM folds beat buy-and-hold.
- GMM has higher fold dispersion, producing both the strongest positive and strongest negative fold among the two models.
- The regime distributions differ substantially, especially for Unknown and Bull, so model comparison is partly a comparison of different exposure timing policies.

Caveats before using the results for decisions

- The report does not independently verify whether transaction costs, bid/ask spreads, slippage or taxes are included in the exported backtest.
- The report does not establish statistical significance of the difference between HMM and GMM.
- The fold construction, hyperparameter selection process and exact model-training procedure are inferred only from the supplied outputs; the underlying training code was not provided as report input.
- Regime labels such as Bull, Bear, Sideways and High-Volatility are descriptive mappings. Their economic meaning depends on how the mapping was defined during model development.
- Historical backtest performance is not a guarantee of future performance. Regime persistence, market structure and execution conditions can change.

Bottom line: The supplied experiment shows a clear trade-off: regime-aware exposure reduced drawdown but did not overcome the long-run return of passive SPY exposure during this evaluation window. Among the two regime models, GMM generated the higher cumulative and annualized strategy return, while HMM had a slightly higher positive-fold rate. The next analytical step is robustness testing under realistic costs, alternative windows, parameter perturbations and statistical significance checks.